THE N-TORSION OF THE PICARD GROUP OF A SMOOTH PROPER CURVE BECOMES THE FULL STANDARD GROUP OVER A BOUNDED SEPARABLE EXTENSION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the bounded rational-point reduction, the two-step torsion descent, the continuous action with open kernel and the factorial Galois bound, and the genus-1 reuse are all retained verbatim.

Editorial note. The label lemma-torsion-picard-becomes-visible names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Rational points over bounded separable extensions. Same chapter, Lemma lemma-point-over-separable-extension; Varieties, Lemma varieties-lemma-smooth-separable-closed-points-dense. Supplies a finite separable extension of degree at most 2g - 2 (g >= 2) or merely finite separable (g = 1) over which X acquires a rational point.
- Torsion over the algebraic closure and its descent. Same chapter, Lemma lemma-torsion-picard-smooth-projective; Picard Schemes of Curves, Lemma pic-lemma-torsion-descends. Identifies Pic(X_{overline{k}})[n] with (Z/nZ)^{2g} and gives the descent of torsion together with the continuous Galois action on Pic(X_{k^{sep}})[n].

Source reference. Original reference 0CD4: sources/stacks-curves/pinned-original-source.tex, source lines 2971–2971. Prerequisite pointer invoked at proof line(s) 3660; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1Z: sources/stacks-curves/pinned-original-source.tex, source lines 3611–3611. Prerequisite pointer invoked at proof line(s) 3666; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CD5: sources/stacks-chow/tags.txt, source lines 15530–15530. Prerequisite pointer invoked at proof line(s) 3670; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 056U: sources/stacks-chow/tags.txt, source lines 6722–6722. Prerequisite pointer invoked at proof line(s) 3696; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let n be prime to the characteristic of k. Let X be a smooth proper curve over k with $H^0(X, \mathcal{O}_X) = k$ and of genus g.*

(1) *If $g = 1$ then there exists a finite separable extension k'/k such that $X_{k'}$ has a k'-rational point and $\mathrm{Pic}(X_{k'})[n] \cong (\mathbf{Z}/n\mathbf{Z})^{\oplus 2}$.*
(2) *If $g \geq 2$ then there exists a finite separable extension k'/k with $[k' : k] \leq (2g-2)(n^{2g})!$ such that $X_{k'}$ has a k'-rational point and $\mathrm{Pic}(X_{k'})[n] \cong (\mathbf{Z}/n\mathbf{Z})^{\oplus 2g}$.*

Proof. Assume $g \geq 2$. First we may choose a finite separable extension of degree at most $2g-2$ such that X acquires a rational point, see Lemma 0CD4. Thus we may assume X has a k-rational point $x \in X(k)$ but now we have to prove the lemma with $[k' : k] \leq (n^{2g})!$. Let $k \subset k^{sep} \subset \overline{k}$ be a separable algebraic closure inside an algebraic closure. By Lemma 0C1Z we have

$$\mathrm{Pic}(X_{\overline{k}})[n] \cong (\mathbf{Z}/n\mathbf{Z})^{\oplus 2g}$$

By Picard Schemes of Curves, Lemma 0CD5 we conclude that

$$\mathrm{Pic}(X_{k^{sep}})[n] \cong (\mathbf{Z}/n\mathbf{Z})^{\oplus 2g}$$

By Picard Schemes of Curves, Lemma 0CD5 there is a continuous action

$$\mathrm{Gal}(k^{sep}/k) \longrightarrow \mathrm{Aut}(\mathrm{Pic}(X_{k^{sep}})[n]$$

and the lemma is true for the fixed field k' of the kernel of this map. The kernel is open because the action is continuous which implies that k'/k is finite. By Galois theory $\mathrm{Gal}(k'/k)$ is the image of the displayed arrow. Since the permutation group of a set of cardinality n^{2g} has cardinality $(n^{2g})!$ we conclude by Galois theory that $[k' : k] \leq (n^{2g})!$. (Of course this proves the lemma with the bound $|\mathrm{GL}_{2g}(\mathbf{Z}/n\mathbf{Z})|$, but all we want here is that there is some bound.)

If the genus is 1, then there is no upper bound on the degree of a finite separable field extension over which X acquires a rational point (details omitted). Still, there is such an extension for example by Varieties, Lemma 056U. The rest of the proof is the same as in the case of $g \geq 2$. □